

Total No. of Questions : 4]

SEAT No. :

PC-57

[Total No. of Pages : 2

[6360]-58

T.E. (Information Technology) (Semester - I)
HUMAN COMPUTER INTERACTION
(2019 Pattern) (314444) (Insem.)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Que 1 or Que 2, Que 3 or Que 4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data, if necessary.

- Q1)** a) Express your opinion on importance of HCI study while designing interface with an example. [5]
- b) Explain any two disciplines contributing to human computer interaction with example. [5]
- c) The psychology of everyday things focuses on feedback and constraint. Illustrate your understanding with an example of product or interface for the following two scenarios - [5]
- i) Product or interface where Feedback should have been provided but currently, it is not there.
 - ii) Product or interface where constraint should not have been provided but currently, it is there.

OR

- Q2)** a) Apply any two concepts of the everyday things by Donald Norman to any application or product of your choice and mention your opinion. [5]
- i) Visible affordances
 - ii) Visible constraints
 - iii) Mapping
 - iv) Causality
 - v) Transfer effects
- b) Describe the process of User centered design principles. [5]
- c) Correlate Microsoft Windows Word application to any two of the following Measurable human factors with proper justification. Measurable human factors: Time to learn, Speed of performance, Rate of user errors, Retention of skills, Subjective satisfaction. [5]

P.T.O.

- Q3)** a) An account holder of a bank visits “Passbook printing kiosk” to print the pending records on the passbook. During the interaction, the user has to select a few options from many, enter account number details. **Categorize** what all **interaction styles** are covered **and** not covered in the above example. [5]
- b) Correlate the relationship between HCI and Ergonomics. [5]
- c) Consider yourself a designer. Which 5 most important **individual differences** will bother you while designing any product or interface for all. [5]

OR

- Q4)** a) Organize the following three scenarios with reference to the fastest to slowest **response time** by human without any disability with right justification. [5]
- i) Touching the base of hot iron.
- ii) Applying brakes to red signal.
- iii) Loud noise of firecracker within proximity to ears.
- b) An interface designer should make the design more intuitive instead of making it ambiguous as it reduces user’s memory load. Discuss any two applications which satisfy the above statement. [5]
- c) Write a long-term memory model script for the following case. Owner went to Veterinary doctor’s clinic along with the dog. [5]



Total No. of Questions : 4]

SEAT No. :

P8580

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Oct-22/TE/Insem-560

T.E. (Information Technology)

HUMAN COMPUTER INTERACTION

(2019 Pattern) (Semester-I) (314444)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4.*
- 2) *Neat Diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicates full marks.*
- 4) *Assume Suitable data if necessary.*

- Q1)** a) List and Explain General Principles of HCI. [5]
b) What is HCI? Explain any two application area of HCI. [5]
c) Express your opinion - "A design should be User-Centric". [5]

OR

- Q2)** a) Why Study of HCI is Important? Explain in details. [5]
b) List and Explain different UCD principles. [5]
c) Explain Psychology of everyday things. [5]

- Q3)** a) "Human emotions play an important role in designing a GUI for any application" Elaborate your answer with example. [5]
b) What is WIMP interface? Explain how to use its elements to design user interface. [5]
c) List five human senses and identify those that are most important to HCI. [5]

OR

- Q4)** a) List different interaction styles. Explain command line interface with advantages and disadvantages. [5]
b) Explain the concept of Ergonomics with example. [5]
c) Differentiate between human short-term memory and long-term memory. [5]



Total No. of Questions : 4]

SEAT No. :

P5057

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[Total No. of Pages : 2

T.E. (Information Technology) (Insem)
HUMAN COMPUTER INTERACTION
(2019 Pattern) (Semester - I) (314444)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** The study of HCI is directly proportional to enhanced user experiences, justify the statement with relevant example. [5]
- b) Identify and explain any three important disciplines contributing to human computer interaction. [5]
- c) List and explain any two of Norman's Principles with relevant example. [5]

OR

- Q2) a)** Correlate Microsoft Windows Operating System to any two of the following. Measurable human factors with proper justification. Measurable human factors: Time to learn, Speed of performance, Rate of user errors, Retention of skills, Subjective satisfaction. [5]
- b) The principles of HCI can be achieved through the following - [5]
- i) Information access interfaces
 - ii) Reducing memory load and
 - iii) Providing alternative interfaces
- Elaborate any two with relevant examples
- c) List and explain any two golden rules of Schneiderman with relevant examples. [5]

P.T.O.

- Q3)** a) Describe and explain structure of memory with diagram. [5]
- i) Sensory
 - ii) Short term memory
 - iii) Long term memory
- b) Write a short note on “Models of interaction”. [5]
- c) Consider yourself an interface designer. Articulate 5 most important individual differences which will bother you while designing any product or interface for all types of humans. [5]

OR

- Q4)** a) “Negative affect can make it harder to do even easy tasks; positive affect can make it easier to do difficult tasks”. Support your opinion with relevant example. [5]
- b) List and explain in short, the two sub-types of Long-Term memory. [5]
- c) Consider yourself an interface designer. What measures will you take to reduce number of “Human errors” committed by users while using the interface? [5]

